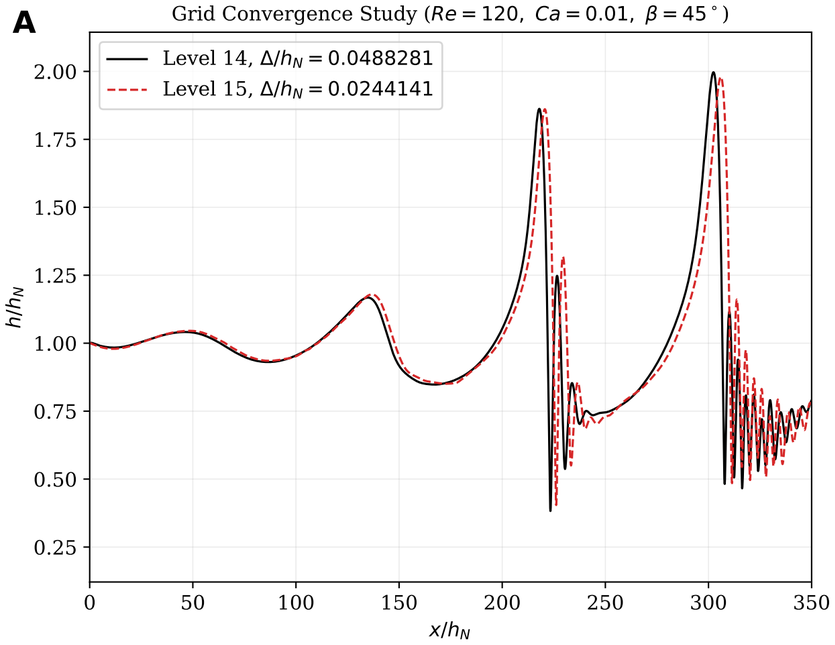
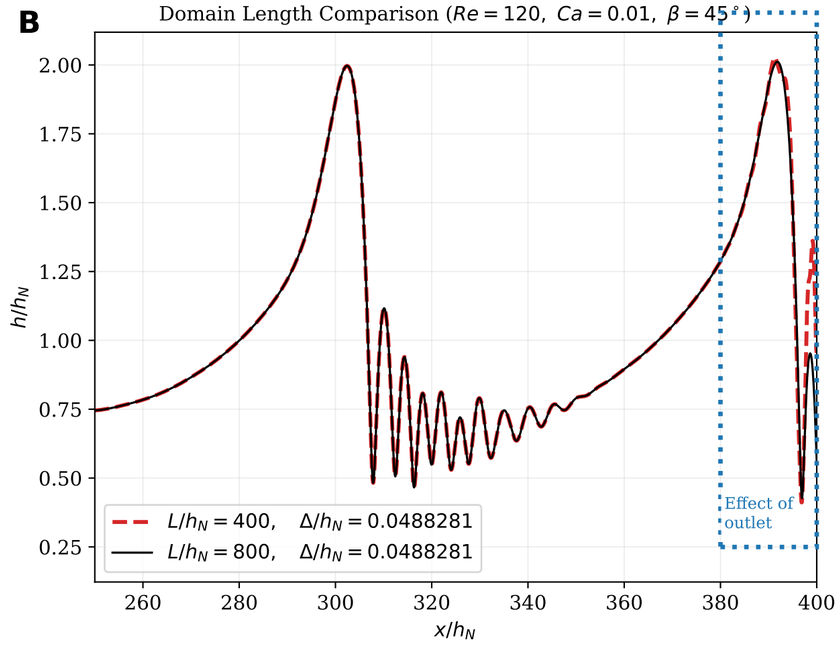
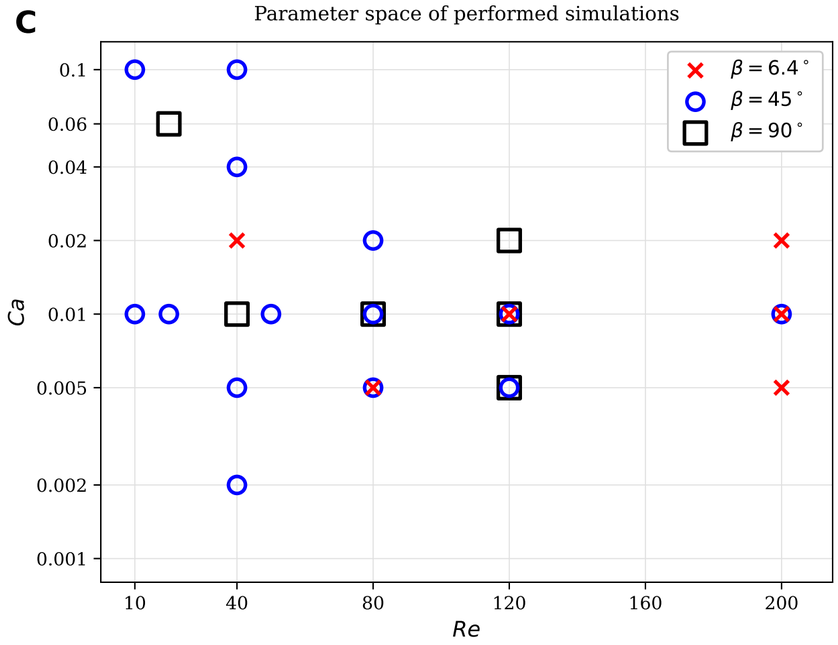
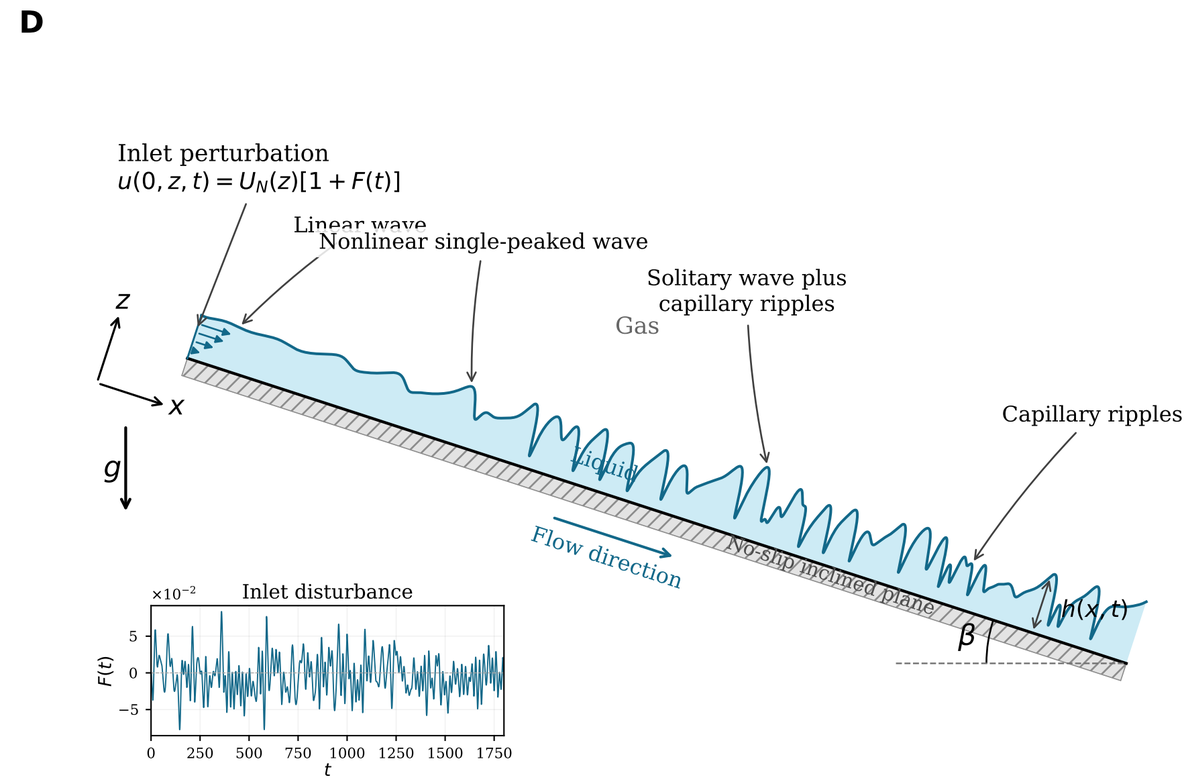
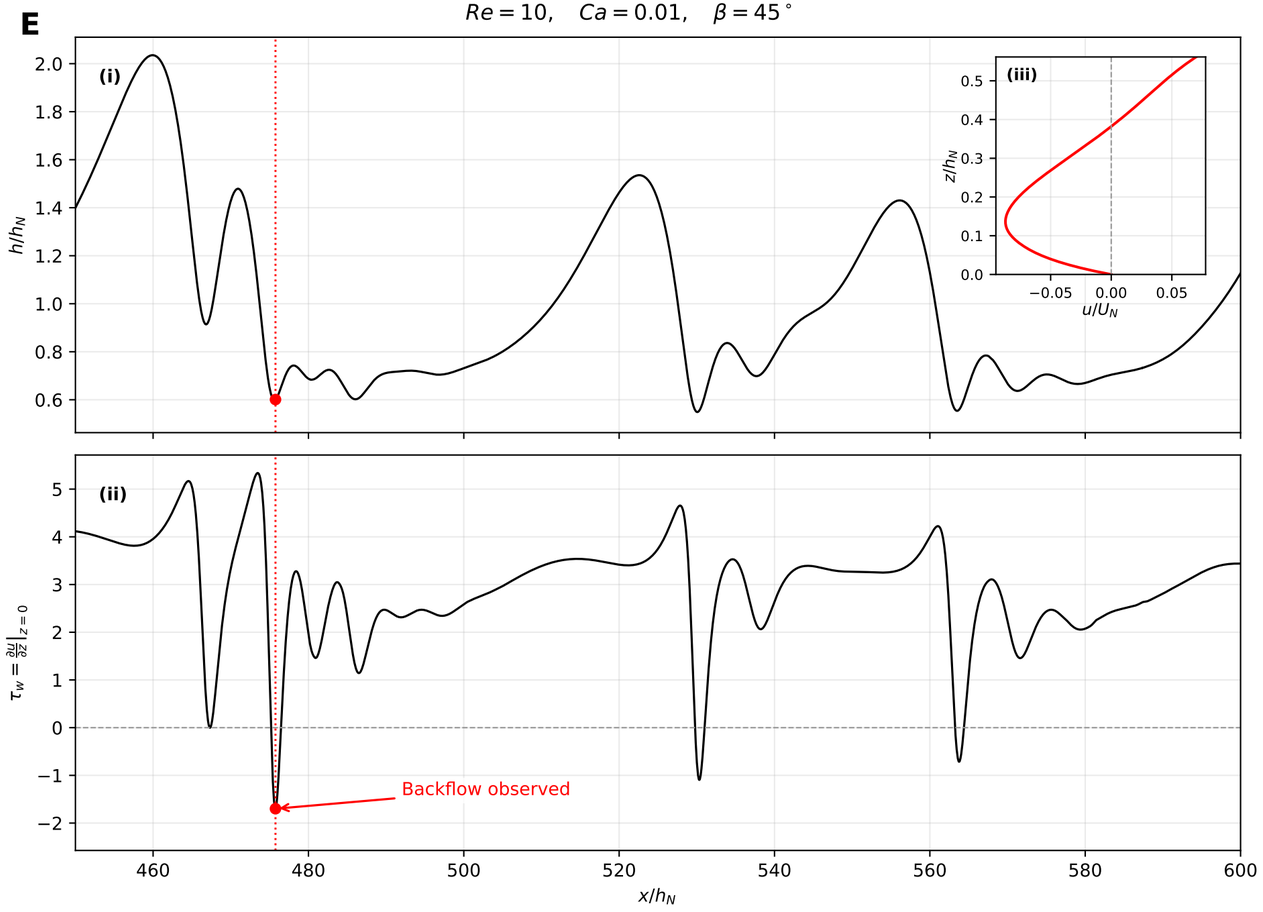

A
Grid Convergence Study ($Re = 120,\ Ca = 0.01,\ \beta = 45^\circ$)
Level 14, $\Delta/h_N = 0.0488281$
Level 15, $\Delta/h_N = 0.0244141$
h/h_N
x/h_N


B
Domain Length Comparison ($Re = 120,\ Ca = 0.01,\ \beta = 45^\circ$)
$L/h_N = 400,\quad \Delta/h_N = 0.0488281$
$L/h_N = 800,\quad \Delta/h_N = 0.0488281$
Effect of outlet
h/h_N
x/h_N


C
Parameter space of performed simulations
$\beta = 6.4^\circ$
$\beta = 45^\circ$
$\beta = 90^\circ$
Ca
Re


D
Inlet perturbation
$u(0, z, t) = U_N(z)[1 + F(t)]$
Linear wave
Nonlinear single-peaked wave
Solitary wave plus capillary ripples
Gas
Capillary ripples
Liquid
Flow direction
No-slip inclined plane
$h(x, t)$
β
z
x
g
Inlet disturbance
$\times 10^{-2}$
$F(t)$
t


E
$Re = 10,\quad Ca = 0.01,\quad \beta = 45^\circ$
(i)
(ii)
(iii)
h/h_N
$\tau_w = \frac{\partial u}{\partial z}\big|_{z=0}$
Backflow observed
x/h_N
z/h_N
u/U_N